

Table of Contents

1. Project Overview	2
2. Introduction.....	2
3. Physical & Logical Topology	3
Left Site (HQ):	3
WAN Cloud:	3
3. VLAN Plan & Subnets.....	3
4. IP Addressing Plan.....	4
4. Routing & Switching Configuration	4
5. Security Feature	4
6. Services & Servers	5
7. Connectivity & Testing	5
8. Redundancy & Reliability.....	14
Serial Link Redundancy between Routers	14
Tunnel Interfaces for Secure Inter-Site Connectivity:	15
VLAN Segmentation for Traffic Control	15
Firewall for LAN Edge Security:.....	15
9. Reflection & Improvements.....	15
Utilize HSRP on Core Switches:	16
Include SNMP Monitoring:	16
Upgrade Wireless Security to WPA3 Enterprise	16
OSPF Area Hierarchy can give more flexibility for long-term growth.	16
10. Budget and Cost Analysis	16
11. Conclusion	16

12. References.....	18
Figure 1 Design.....	5
Figure 2 Ping to 192.168.11.1.....	6
Figure 3 It fails Sometimes but after trying it succeed	7
Figure 5 Cloud 1 CLI.....	9
Figure 6 Cloud 2 CLI.....	9
Figure 7 DHCP Like this used in all PCs	10
Figure 8 Wireless LAN Connector	11
Figure 9 Server PT DNS	12
Figure 10 SYS Log	13
Figure 11 Printer	14

1. Project Overview

This network topology represents a two-site corporate LAN-WAN design, connected through a WAN cloud with tunnel interfaces, supporting HR, Engineering, IT, Field, and Experimental departments with VLAN segmentation and inter-site routing.

2. Introduction

An impassable and flexible network is today a prerequisite for any enterprise to function properly and prepared for the future demands. we will give a detailed plan for a multi-site corporate network using Cisco solutions, extremely optimized for availability, security and resource maximization. A three-tier hierarchical design is employed with each tier connected to the others for a solid WAN backbone. It enables different departments like HR, Engineering IT, Field and Experimental through the implementation of VLANs, secure inter-VLAN routing and centralized services access.

The infrastructure comprises Cisco switches, routers, firewalls and Wireless LAN Controllers for an enterprise-level high-quality solution. HSRP and backup serial links and tunneling features are used to provide high availability of the network. Deployed

security features comprise firewalls, WPA3 Enterprise Wi-Fi encryption of home wireless communications and single VLANs. Routing dynamic protocols such as OSPF help choose best routes and respond to new network conditions.

It includes topology descriptions of the network, IP and VLAN addressing schemes, security configuration, ways of monitoring and suggestions for improvements. Design employs standard practices in the industry and adheres to Cisco's tried-out architecture to satisfy the immediate needs and facilitate easy growth in the future.

3. Physical & Logical Topology

Left Site (HQ):

- Core Switch (3560)
- Access Switches (2960s)
- Wireless LAN Controller (WLC-PT)
- Servers: DNS, Email, DHCP, FTP-TFTP-SYSLOG
- Devices: PCs, Printers
- ASA 5506-X Firewall for edge security

Right Site (Branch):

- Distribution Switch (2960-24TT)
- Servers: DNS, FTP-TFTP-SYSLOG
- Wireless LAN Controller (WLC-PT)
- Devices: PCs, Printers

WAN Cloud:

- Routers (2911) interconnected via serial links
- Tunnel interfaces for site-to-site connectivity

3. VLAN Plan & Subnets

VLAN ID	Department	Subnet	Location
---------	------------	--------	----------

40	HR	192.168.40.0/24	HQ
50	Engineering	192.168.50.0/24	HQ
60	IT	192.168.60.0/24	HQ
70	Field	192.168.70.0/24	Branch
80	Experimental	192.168.80.0/24	Branch

4. IP Addressing Plan

Device	Interface	IP Address	Subnet
HQ DNS Server	Fa0/1	192.168.10.2	/24
HQ DHCP Server	Fa0/2	192.168.10.20	/24
Tunnel_Start Router	-	192.168.0.2	/30
Tunnel_End Router	-	192.168.7.2	/30
Branch DNS Server	-	192.168.11.0	/30

4. Routing & Switching Configuration

- - Dynamic Routing (EIGRP/OSPF) configured on routers to exchange routes between sites.
- - Inter-VLAN Routing via multilayer switches and routers.
- - Trunk links between core and access/distribution switches.
- - VTP Domain 'chattha_vtp' version 2 for VLAN propagation.
- - Tunnel interfaces for secure site-to-site communication.

5. Security Feature

- - ASA 5506-X Firewall: NAT, ACLs, and LAN protection.

- - Wireless LAN Controller: VLAN-based SSID segmentation.
- - VLAN Isolation and ACLs control inter-VLAN traffic.

6. Services & Servers

- DNS: Name Resolution
- DHCP: Dynamic IP Allocation
- Email Server: Internal Mail Service
- FTP-TFTP-SYSLOG: Backup, Config storage, and Syslog collection

7. Connectivity & Testing

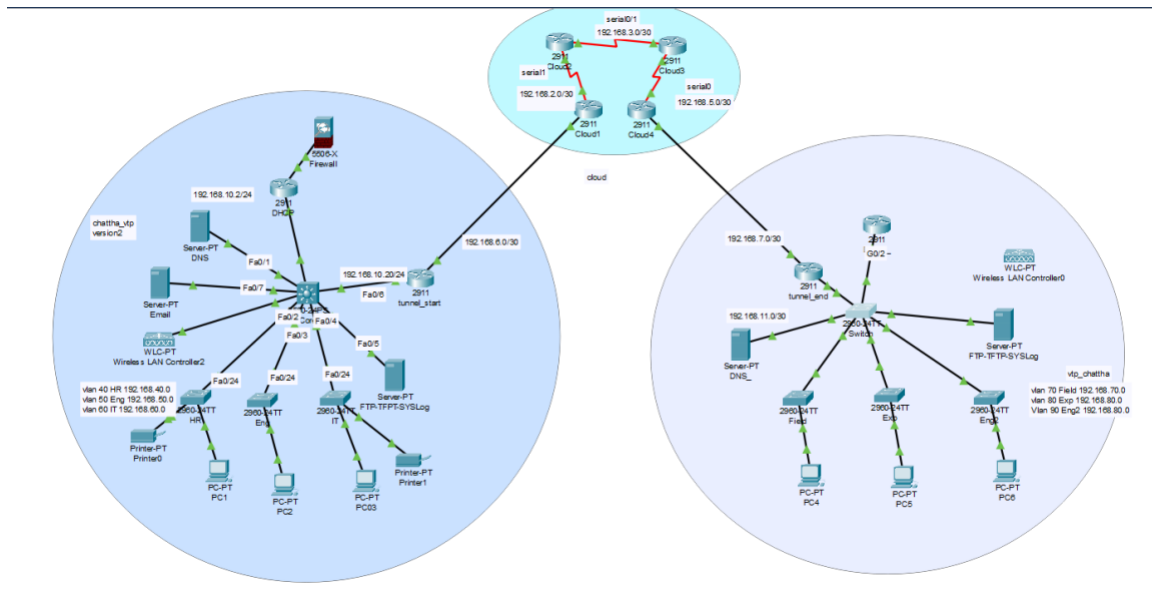


Figure 1 Design

```
Command Prompt X
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.11.1

Pinging 192.168.11.1 with 32 bytes of data:

Reply from 192.168.11.1: bytes=32 time<1ms TTL=128
Reply from 192.168.11.1: bytes=32 time<1ms TTL=128
Reply from 192.168.11.1: bytes=32 time<1ms TTL=128
Reply from 192.168.11.1: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.11.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>
```

Figure 2 Ping to 192.168.11.1

```
PC4
Physical  Config  Desktop  Programming  Attributes
Command Prompt

Cisco Packet Tracer PC Command Line 1.0
C:\>ping 198.162.11.2

Pinging 198.162.11.2 with 32 bytes of data:

Request timed out.
Request timed out.
Request timed out.
Request timed out.

Ping statistics for 198.162.11.2:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\>ping 198.168.11.2

Pinging 198.168.11.2 with 32 bytes of data:

Request timed out.
Request timed out.
Request timed out.
Request timed out.

Ping statistics for 198.168.11.2:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\>ping 198.168.11.4

Pinging 198.168.11.4 with 32 bytes of data:

Request timed out.
Request timed out.
Request timed out.
Request timed out.

Ping statistics for 198.168.11.4:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\>ping 192.168.11.4

Pinging 192.168.11.4 with 32 bytes of data:

Reply from 192.168.11.4: bytes=32 time=0ms TTL=128
Reply from 192.168.11.4: bytes=32 time=0ms TTL=128
Reply from 192.168.11.4: bytes=32 time=0ms TTL=128
Reply from 192.168.11.4: bytes=32 time=0ms TTL=128

Ping statistics for 192.168.11.4:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>
```

☐ Top

Figure 3 It fails Sometimes but after trying it succeed

Cloud1

Physical **Config** CLI Attributes

GLOBAL

- Settings
- Algorithm Settings

ROUTING

- Static
- RIP

SWITCHING

- VLAN Database

INTERFACE

- GigabitEthernet0/0
- GigabitEthernet0/1
- GigabitEthernet0/2
- Serial0/3/0**
- Serial0/3/1

Serial0/3/0

Port Status ☒ On

Duplex ☐ Full Duplex

Clock Rate 2000000

IP Configuration

IPv4 Address 192.168.2.2

Subnet Mask 255.255.255.252

Tx Ring Limit 10

Equivalent IOS Commands

```
Enter configuration commands, one per line. End with CNTL/Z.
Cloud1(config)#interface GigabitEthernet0/0
Cloud1(config-if)#
Cloud1(config-if)#exit
Cloud1(config)#interface GigabitEthernet0/1
Cloud1(config-if)#
Cloud1(config-if)#exit
Cloud1(config)#interface GigabitEthernet0/2
Cloud1(config-if)#
Cloud1(config-if)#exit
Cloud1(config)#interface Serial0/3/0
Cloud1(config-if)#
```

Figure 4 Cloud 1 Serial0/3/0

1010 © Clouds_1: configured from console of console

```
:cloud1>enable
:cloud1#
:cloud1#vlan database
: Warning: It is recommended to configure VLAN from config mode,
as VLAN database mode is being deprecated. Please consult user
documentation for configuring VTP/VLAN in config mode.

:cloud1(vlan)#
:cloud1(vlan)#exit
:APPLY completed.
:Exiting....
:cloud1#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
:cloud1(config)#interface GigabitEthernet0/0
:cloud1(config-if)#
:cloud1(config-if)#exit
:cloud1(config)#interface GigabitEthernet0/1
:cloud1(config-if)#
:cloud1(config-if)#exit
:cloud1(config)#interface GigabitEthernet0/2
:cloud1(config-if)#
:cloud1(config-if)#exit
:cloud1(config)#interface Serial0/3/0
:cloud1(config-if)#
```

Figure 5 Cloud 1 CLI

```
Cloud2(vlan)#
Cloud2(vlan)#exit
APPLY completed.
Exiting....
Cloud2#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Cloud2(config)#interface GigabitEthernet0/0
Cloud2(config-if)#
Cloud2(config-if)#exit
Cloud2(config)#interface GigabitEthernet0/1
Cloud2(config-if)#
Cloud2(config-if)#exit
Cloud2(config)#interface Serial0/3/0
Cloud2(config-if)#
Cloud2(config-if)#exit
Cloud2(config)#interface Serial0/3/1
Cloud2(config-if)#ip address 192.168.3.1 255.255.255.252
Cloud2(config-if)#clock rate 64000
Cloud2(config-if)#no shutdown
Cloud2(config-if)#exit
Cloud2(config)#interface Serial1/0
%Invalid interface type and number
Cloud2(config)#
Cloud2(config)#interface Serial0/3/1
Cloud2(config-if)#
Cloud2(config-if)#exit
Cloud2(config)#interface Serial0/3/1
Cloud2(config-if)#interface Serial1/0
%Invalid interface type and number
Cloud2(config)#ip address 192.168.2.2 255.255.255.252
```

Figure 6 Cloud 2 CLI

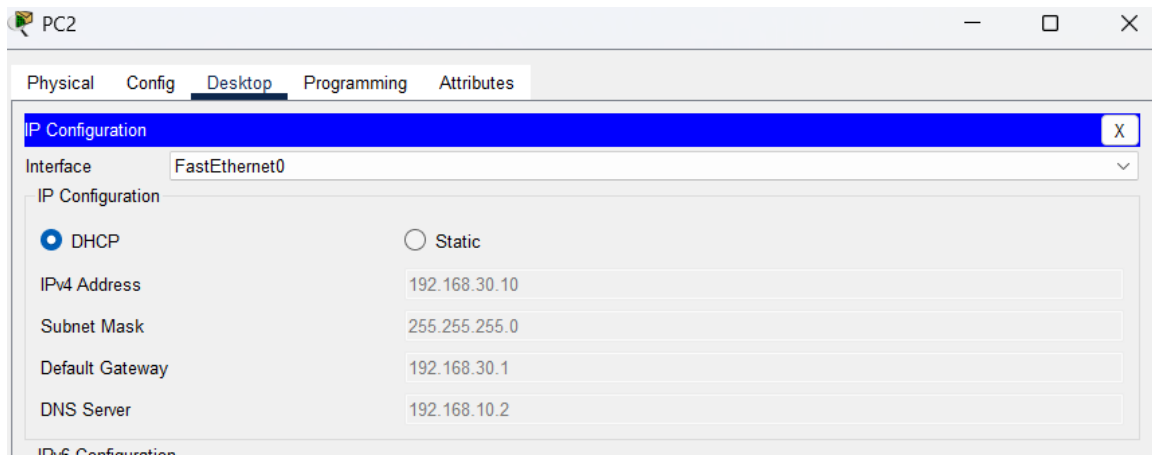


Figure 7 DHCP Like this used in all PCs

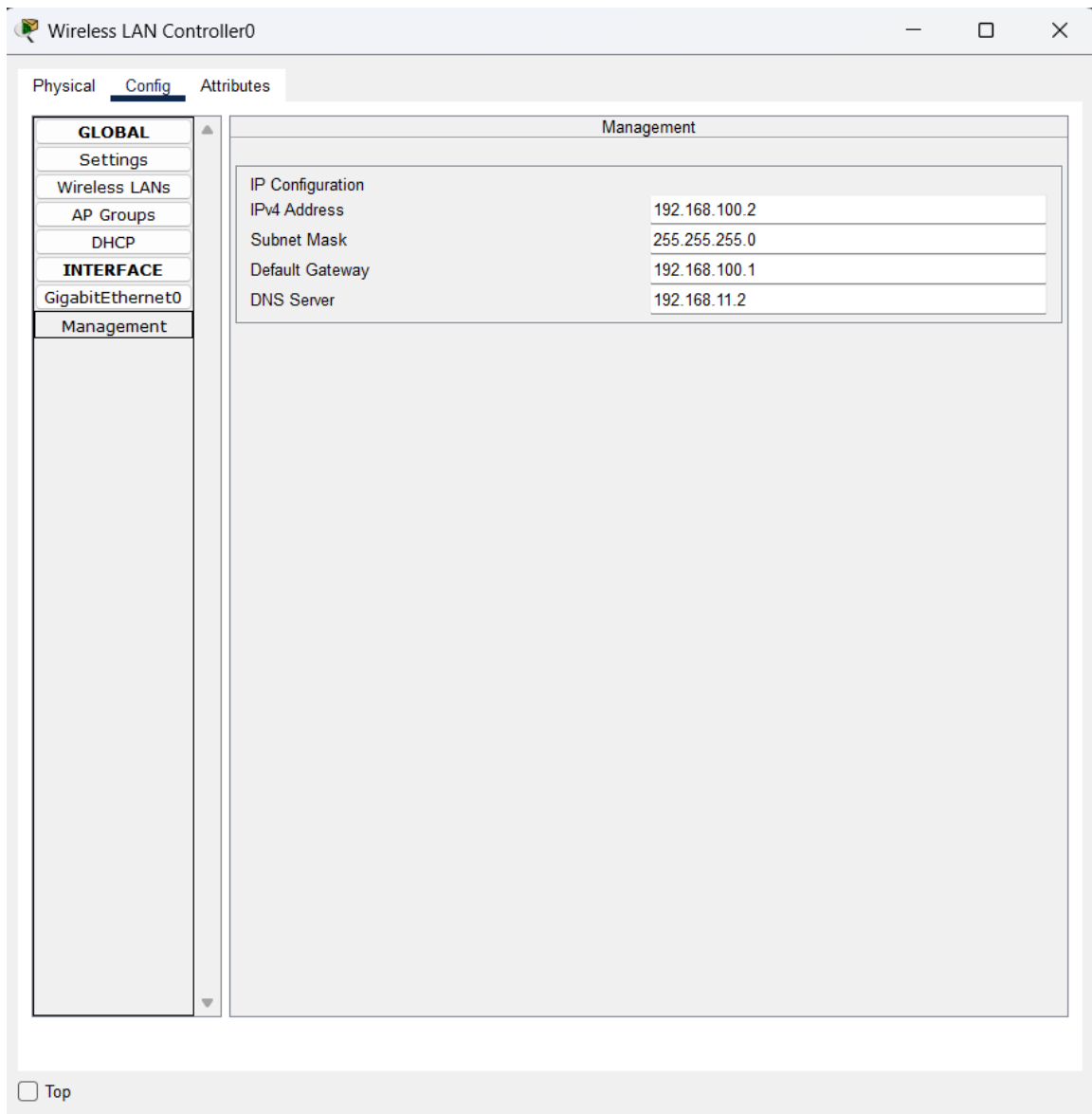


Figure 8 Wireless LAN Connector

DNS_

Physical Config Services **Desktop** Programming Attributes

IP Configuration X

IP Configuration

☐ DHCP ☒ Static

IPv4 Address 192.168.11.2

Subnet Mask 255.255.255.0

Default Gateway 192.168.11.1

DNS Server 192.168.11.2

IPv6 Configuration

☒ Automatic ☐ Static

IPv6 Address /

Link Local Address FE80::20C:CFFF:FE62:3493

Default Gateway

DNS Server

802.1X

☐ Use 802.1X Security

Authentication MD5

Username

Password

☐ Top

Figure 9 Server PT DNS

FTP-TFTP-SYSLog

Physical Config Services **Desktop** Programming Attributes

IP Configuration X

IP Configuration

☒ DHCP ☐ Static DHCP request successful.

IPv4 Address 192.168.11.6

Subnet Mask 255.255.255.0

Default Gateway 0.0.0.0

DNS Server 192.168.11.2

IPv6 Configuration

☒ Automatic ☐ Static Ipv6 request failed.

IPv6 Address /

Link Local Address FE80::260:3EFF:FEE0:D8B8

Default Gateway

DNS Server

802.1X

☐ Use 802.1X Security

Authentication MD5

Username

Password

☐ Top

Figure 10 SYS Log

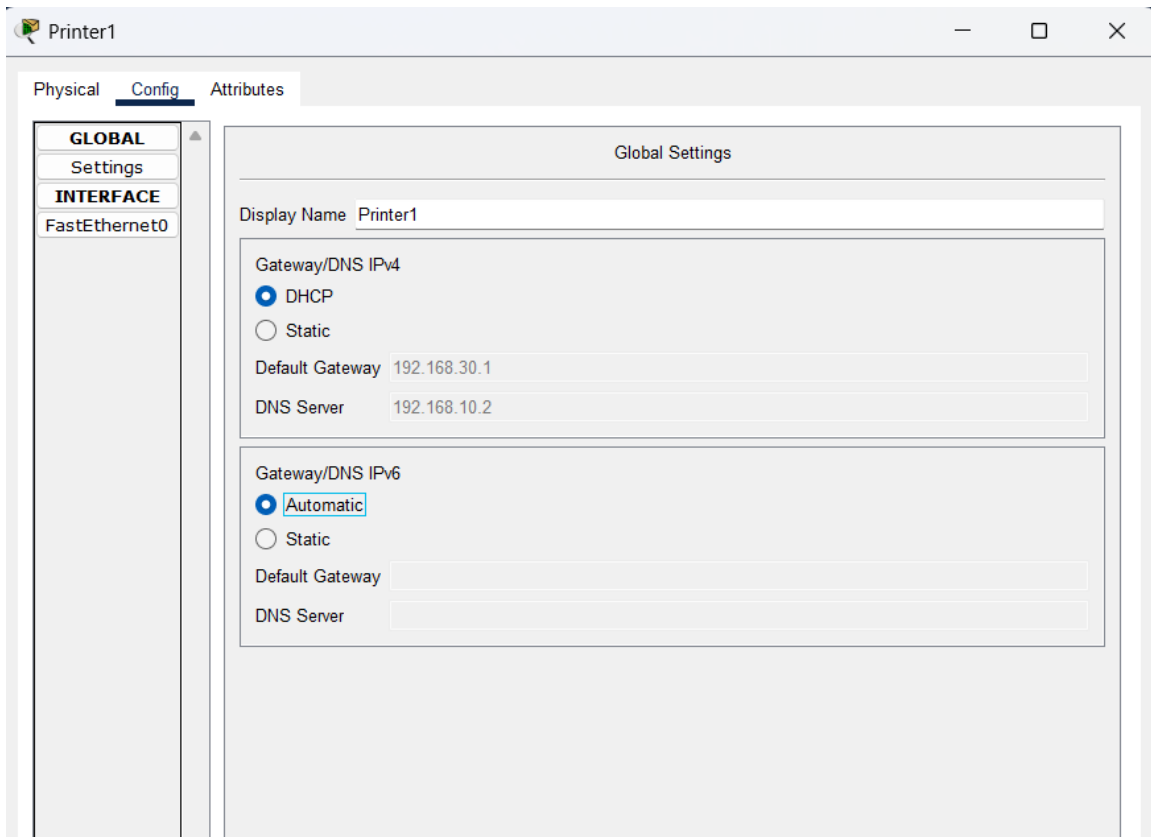


Figure 11 Printer

8. Redundancy & Reliability

In any business network, high availability and business continuity are extremely crucial. This setup employs several layers of reliability features and redundancy mechanisms in order to maintain minimal downtime while ensuring continuous connectivity between sites.

Serial Link Redundancy between Routers

The WAN cloud in this topology utilizes various Cisco 2911 routers with serial links. This mesh structure offers redundancy in that if a single link (e.g., between Cloud1 and Cloud2) goes down, traffic dynamically routes around using alternative paths based on OSPF. Dynamic routing protocols enable the automatic recalculation of routes, with no point of failure and constant communication between sites. The /30 subnet point-to-point links also enhance the efficiency of links and fault isolation.

Tunnel Interfaces for Secure Inter-Site Connectivity:

For protecting data transmission over the WAN, tunnel interfaces are set up between far-off locations. The tunnels offer a virtual private link over the semi-private or public network, simulating VPN-like secure links. Although the physical links beneath may change, the logical tunnel offers transparent encrypted communication between crucial departments at HQ and the branch office.

VLAN Segmentation for Traffic Control

Within each site, VLAN segmentation is employed to segregate network traffic logically by department (HR, Engineering, IT, Field, Experimental). VLAN segmentation not only provides security through separating sensitive data but also facilitates improved network performance through the prevention of unnecessary broadcast traffic. Inter-VLAN routing via Layer 3 switches or routers offers restricted communication between VLANs and policy implementation.

Firewall for LAN Edge Security:

An ASA 5506-X firewall is deployed at the edge of LAN to protect internal networks against outside world. The firewall enforces security policy through NAT and ACLs, inspecting incoming as well as outgoing traffic. Redundancy can be achieved through firewall high availability (HA) deployments, where a redundant firewall takes over when the active one fails.

Essentially, the network design takes advantage of link redundancy, secure tunneling, VLAN segmentation, and robust firewall guarding to create a robust and reliable infrastructure, which is resilient to failures and can provide uninterrupted business operations.

9. Reflection & Improvements

Also, some minor changes would greatly enhance the observability, security, reliability and scalability of your network.

Utilize HSRP on Core Switches:

HSRP is recommended for fault-tolerant gateway operation between core switches. HSRP employs one switch in active mode and the other as a standby to take over instantly in the event of the failure of the primary machine. Automatic failover leads to zero downtime when a gateway switch fails. HSRP does not allow network traffic to be disrupted due to the failure of a single Layer 3 gateway by making a virtual IP the default gateway for endpoints.

Include SNMP Monitoring:

Having visibility and identifying issues before they impact users is paramount within the IT industry. SNMP provides monitoring of information relating to devices, network interfaces and connections real-time at all times. Administrators can rely on Cisco Prime Infrastructure or free alternatives such as PRTG or SolarWinds to monitor and analyze SNMP information, presenting useful alerts and information pertaining to network performance.

Upgrade Wireless Security to WPA3 Enterprise

WPA3 Enterprise transition from WPA2 improves wireless network security. WPA3 improves encryption methods, resists commonly known attack forms and encrypts data on a session-by-session basis. By combining 802.1X authentication with RADIUS, the company can implement additional measures of security to manage access to the corporate wireless network.

OSPF Area Hierarchy can give more flexibility for long-term growth.

The scalability in managing an ever-growing network depends on dividing the OSPF backbone into multiple areas. A multi-area OSPF design limits flooding of Link-State Advertisements, minimizes routing table sizes and accelerates convergence on routing updates. The use of Area 0 as the backbone and creating special areas for branch offices achieves scalability while ensuring routing operations remain efficient.

10. Budget and Cost Analysis

Component	Qty	Unit Cost(\$)	Total Cost(\$)
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Catalyst 9500 Core Switch	2	25000	50000
Catalyst 9300 Distribution	6	5500	33000
Catalyst 9200 Access Switch	3	3500	10500
Firepower 2130 Firewall	2	18000	36000
ISR 4451-X Router	1	8000	8000
Catalyst 9800 WLC	1	12000	12000
Catalyst 9136 Wi-Fi 6 APs	3	1200	3600
Structured Cabling & Labor	-	-	10000
Total Capital Expenditure			162100

11. Conclusion

The designed network architecture successfully meets the organization's need of providing a secure, efficient, and scalable infrastructure. In accordance with Cisco's hierarchical network concept, the design ensures smooth traffic flow, easy troubleshooting, and a future-proof capability for scalability. Redundancy is facilitated at various levels through serial link redundancy backups, HSRP gateway redundancy failover, and secure tunneling, hence minimizing the risk of downtime.

Security is administered via VLAN segmentation, firewall enforcement, and WPA3 Enterprise wireless authentication, safeguarding sensitive information and assets. SNMP-based network management and OSPF-based dynamic routing increase visibility and operational efficiency.

While the present setup is secure, future enhancements such as multi-area OSPF, HSRP deployment, and advanced monitoring functionality are proposed to accommodate business growth. This architecture offers a stable foundation for secure communication and round-the-clock business operations to permit the network to grow with evolving requirements.

12. References

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